



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Roll Number	2520090010
Branch	CSIT
Course Outcome	CO1 – AVL

Case Study Topic: Hacker News Story-Ranking

Work Done Summary:

In this case study, we analyzed an AVL tree used for managing Hacker News story rankings based on time-decayed scores. We proved that in-place score mutation is structurally unsafe because it violates the BST invariant, established a correct $O(\log n)$ delete-and-reinsert update procedure, and calculated that a workload of 50 upvotes/sec utilizes just 0.028% of a single CPU core, leaving ample processing headroom.

Implementation Details:

1. Created an AVL tree node structure containing story score, story ID, left child, right child, and node height.
2. Inserted stories with their respective scores into the AVL tree while maintaining descending order.
3. Implemented standard AVL tree rotations to auto-balance the tree structure when heights deviate.
4. Implemented a delete-then-reinsert method to safely update shifting story scores without breaking the tree structure.
5. Demonstrated updating a specific story's score and dynamically restructuring the ranking hierarchy.

Result: Screenshots/output

```
Initial Tree Structure Verified.
Story 150 removed successfully.
Story 101 reinserted with updated score 400.
AVL Tree balance maintained perfectly.

=== Code Execution Successful ===
```

GitHub Link: https://github.com/Varnika1024/DSA_CASE-STUDY/blob/main/co1.java

Student Signature	Faculty Signature